

Problem 11.6

An investment of \$500 was made three years ago. For the first six months, the annual effective rate of interest paid on the account was 5%, but it then increased to 7% and did not change over the following two and a half years. Find the present accumulated value of the investment.

Solution

The investment earns an annual effective rate of 5% for the first half-year. Its value after this period is therefore

$$500(1.05)^{1/2}.$$

It then earns an annual effective rate of 7% for the remaining two and a half years. Thus, the present accumulated value is

$$\begin{aligned} A &= 500(1.05)^{1/2}(1.07)^{5/2} \\ &= 500(1.05)^{0.5}(1.07)^{2.5} \\ &\approx 606.769997. \end{aligned}$$

Therefore, the present accumulated value of the investment is

$$\boxed{A \approx \$606.77}.$$